Chapter 4: Multi-dimensional Motion

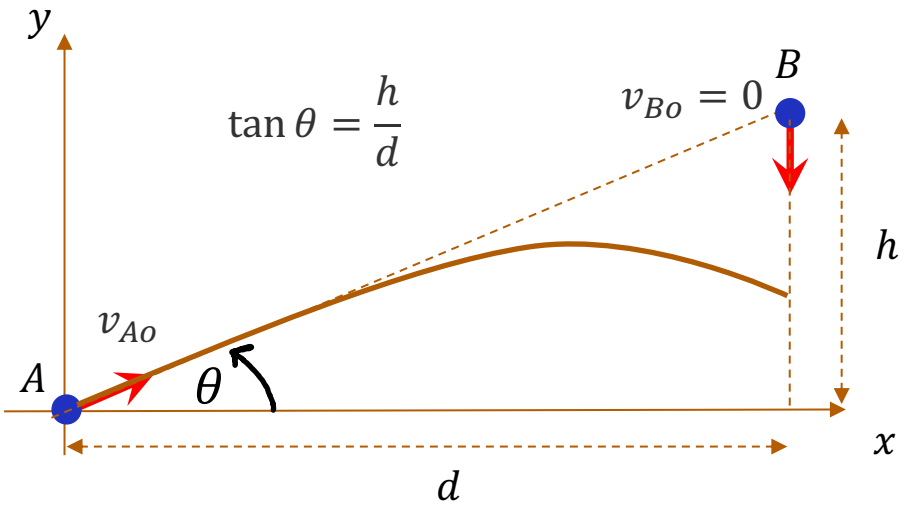
(Constant Acceleration)

The following subjects will be discussed

- Equations of motion in 3D.
- Examples.
- Special cases.

Example

At what conditions will the balls collide mid-air?



Ball A: Projectile motion

$$x = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$d = v_{Aox}t$$

$$t = \frac{d}{v_{Aox}}$$

$$y = y_o + v_{oy}t - \frac{1}{2}gt^2$$

$$y_A = v_{Aoy}\frac{d}{v_{Aox}} - \frac{1}{2}gt^2$$

$$y_A = h - \frac{1}{2}gt^2$$

Ball B: Free-fall

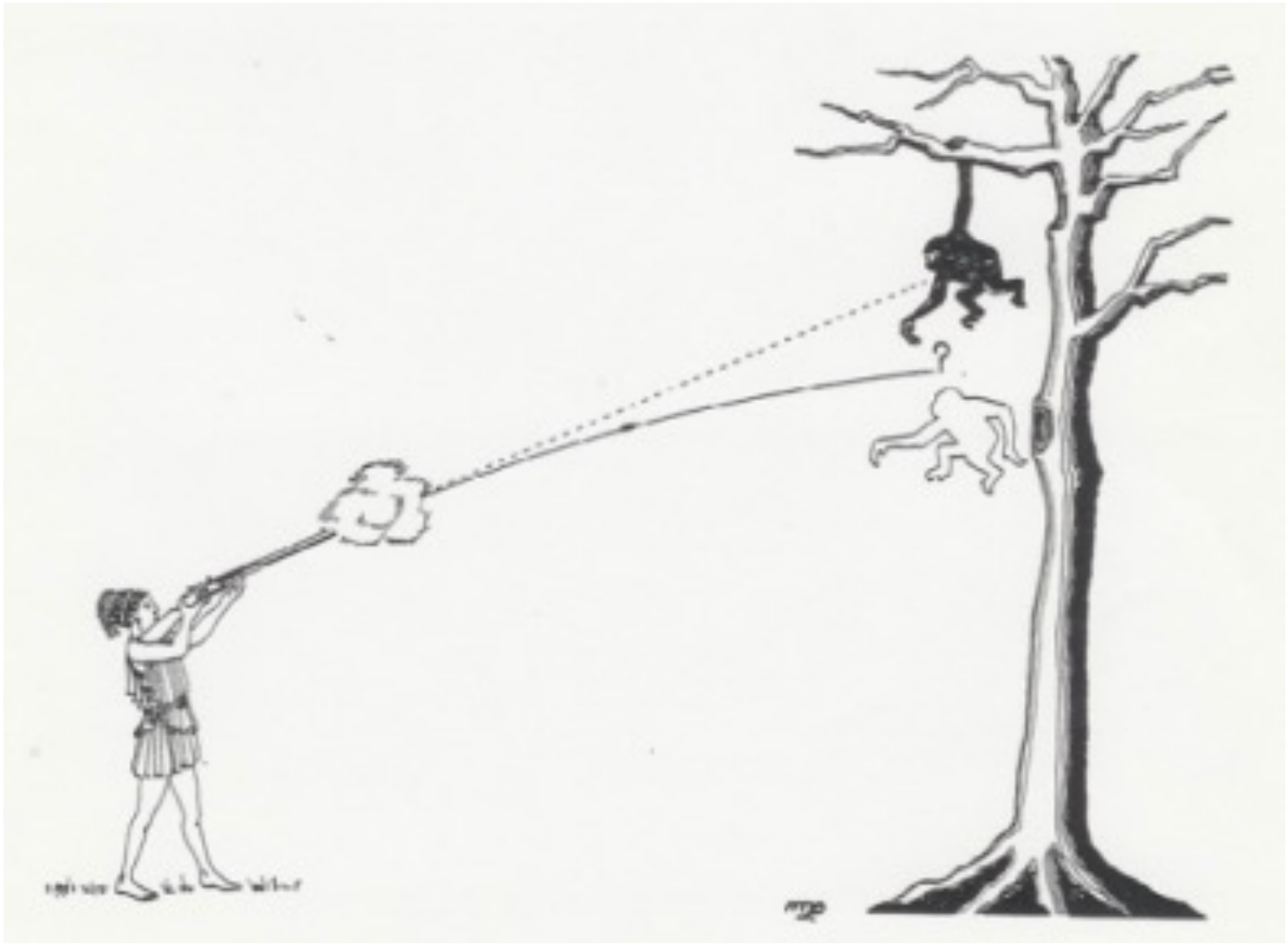
$$y = y_o + v_{oy}t - \frac{1}{2}gt^2$$

$$y_B = h - \frac{1}{2}gt^2$$

Conclusion

Balls will collide for any initial speed of A, as long as A is initially aimed at B.

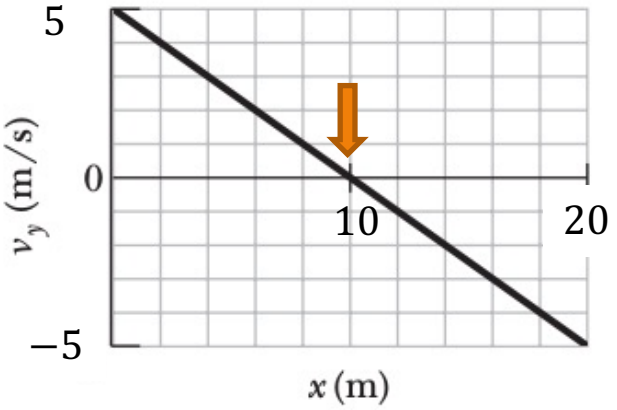
A (Sad) Example



Example

Find the angle to hit the bottom of the wall.





$$v_{oy} = 5\ m/s$$

$$v_o = \frac{v_{oy}}{\sin\theta} = \frac{5}{\sin\theta}$$

$$x = 10 = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$10 = v_o \cos\theta \cdot t$$

$$10 = \frac{5}{\sin\theta}\cos\theta \cdot t \qquad t = 2\tan\theta$$

$$v_y = 0 = v_{oy} - g\,t$$

$$5 = g\ 2\tan\theta$$

$$\tan\theta = \frac{5}{2g}$$

$$\theta = 14°$$

Example

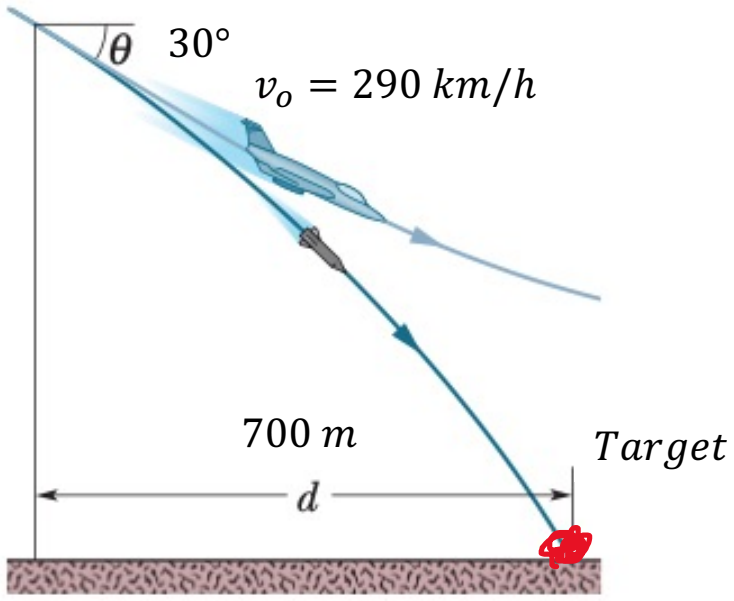
How long does the missile take to hit target?

$$v_{ox} = v_o \cos\theta = 290\cos(-30) = 69.8\ m/s$$

$$v_{oy} = v_o \sin\theta = 290\sin(-30) = -40.3\ m/s$$

$$x = 700 = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$700 = v_{ox}t \qquad t = \frac{700}{69.8} = 10\ s$$



How high was the release point?

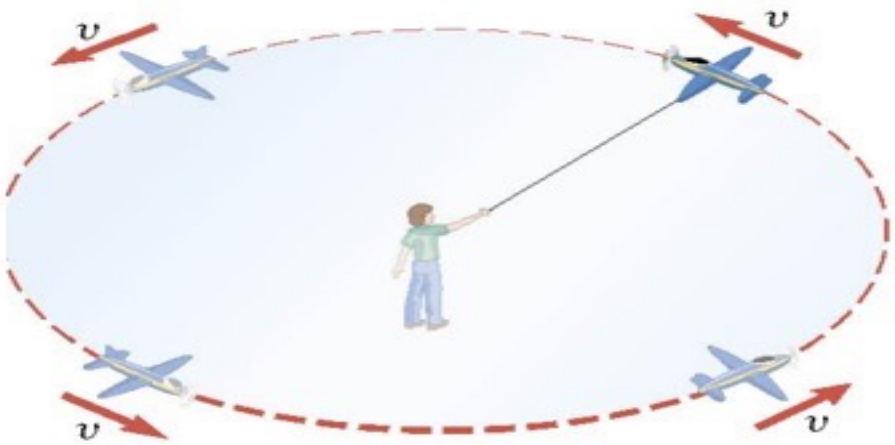
$$y - y_o = v_{oy} \cdot t - \frac{1}{2} g\, t^2$$

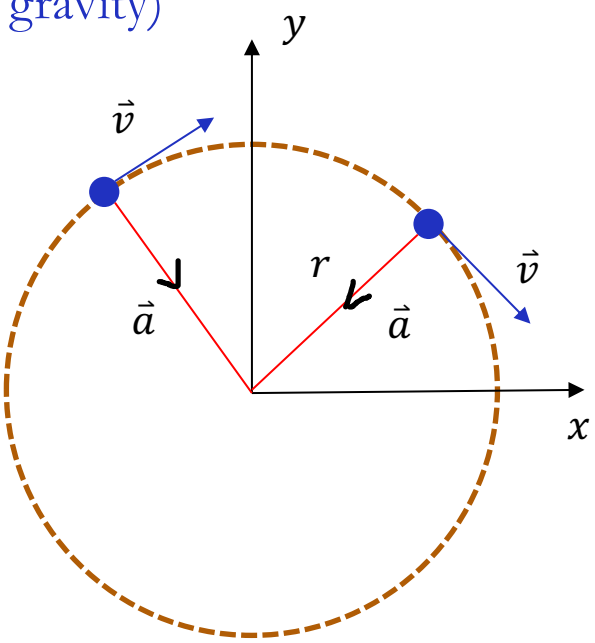
$$-y_o = (-40.3)(10) - \frac{1}{2}(9.8)\,(10)^2$$

$$y_o = 893\ m$$

Uniform Circular Motion

A 2D motion in a horizontal plane (no gravity)





Velocity magnitude is constant

Velocity direction is continuously changing

So, the acceleration is NOT zero

$$\vec{a} = \frac{v^2}{r}\,\vec{a}_n$$

Time for one cycle (Period)

$$T = \frac{2\pi r}{v}$$

Example

Consider an object in counterclockwise circular motion with constant velocity

At $t_1 = 2.00\ s,$ $\vec{a} = 6\,\vec{i} + 4\,\vec{j}$

At $t_2 = 5.00\ s,$ $\vec{a} = 4\,\vec{i} - 6\,\vec{j}$

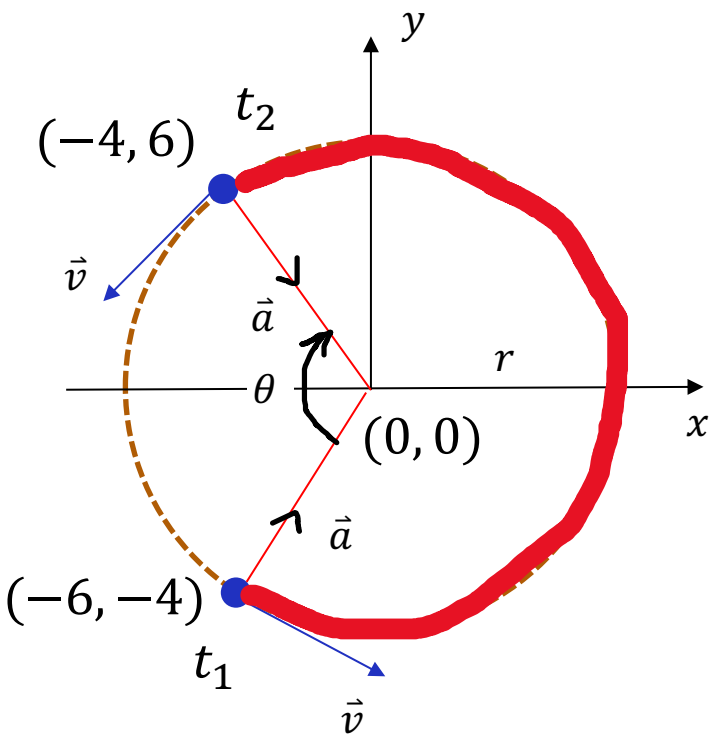
Find the radius.

The velocity is arc length/time.

$$\theta = \tan^{-1}\frac{6}{4} + \tan^{-1}\frac{4}{6} = 1.57\ rad$$

Arc length=(2π-1.57)r

$$v = \frac{arc}{t} = \frac{4.71\ r}{5-2} = 1.57\ r$$

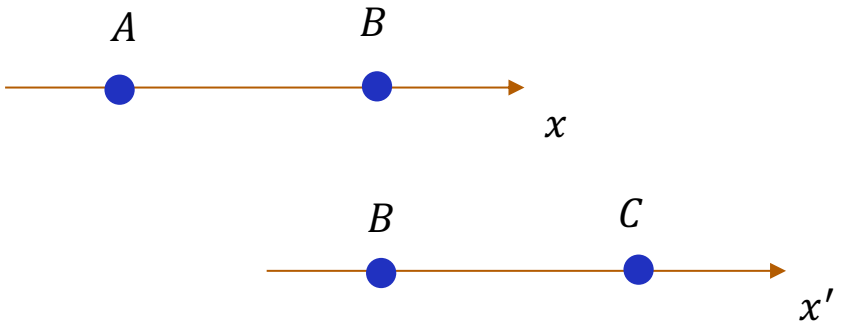


$$|a| = 7.2 = \frac{v^2}{r} = \frac{(1.57\ r)^2}{r}$$

$$r = 2.92\ m$$

Relative Motion

Different frame of references can be defined and used. The idea is to relocate the origin to make the motion analysis easier.

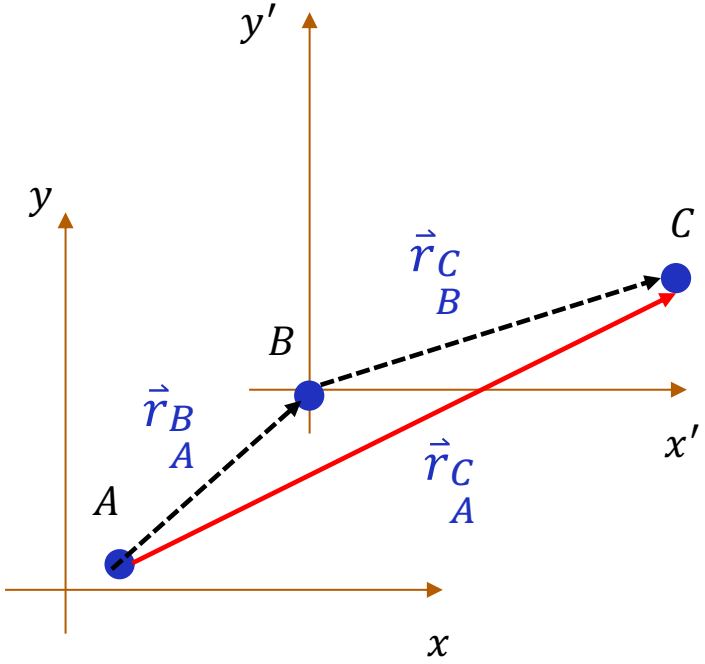


v_B Constant

$$x_{C/A} = x_{B/A} + x_{C/B}$$

$$v_{C/A} = v_{B/A} + v_{C/B}$$

$$a_{C/A} = a_{C/B}$$



$$\vec{r}_{C/A} = \vec{r}_{B/A} + \vec{r}_{C/B}$$

$$\vec{v}_{C/A} = \vec{v}_{B/A} + \vec{v}_{C/B}$$

$$\vec{a}_{C/A} = \vec{a}_{C/B}$$

Example

Find the person's speed to the sidewalk speed ratio.

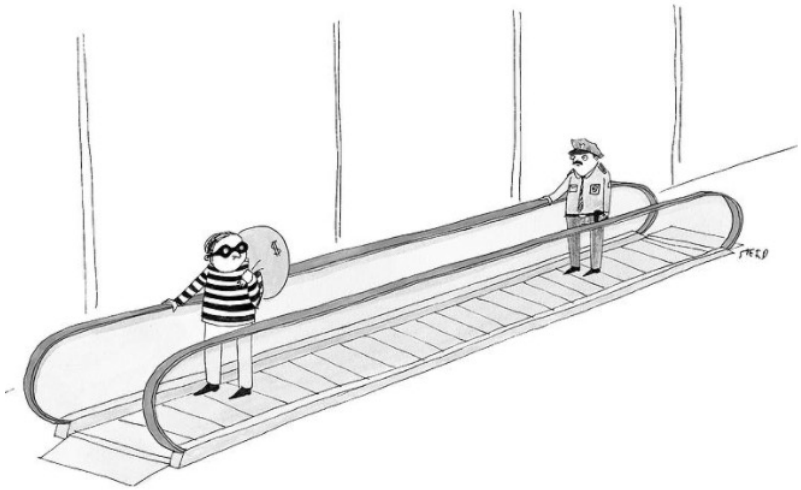
$$v_{P/S} \; / v_{S/A}$$

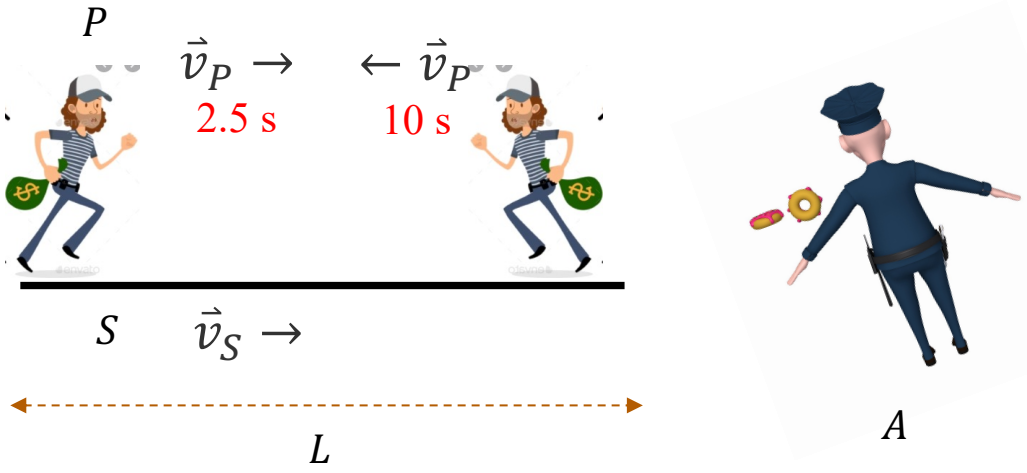
$$\rightarrow v_{P/A} = v_{S/A} + v_{P/S} = \frac{L}{2.5}$$

$$\leftarrow v_{P/A} = v_{S/A} - v_{P/S} = \frac{-L}{10}$$

$$v_{S/A} + v_{P/S} + 4v_{S/A} - 4v_{P/S} = 0$$

$$\frac{v_{P/S}}{v_{S/A}} = 5/3$$





Chapter 5: Force and Motion

The following subjects will be discussed

- Newton's Laws of motion.
- Types of Forces
- Examples.

Introduction

Forces in nature. Non-contact forces.

Forces in this course. Contact forces.

From Galileo to Newton. Newtonian motion. Relativistic motion.

Quantum Mechanical dynamics.

Newton's First Law

The "motional" status of an object stays the same if no net external forces is acting on it

Newton's Second Law

If there is a net external force acting on an object, it will accelerate.

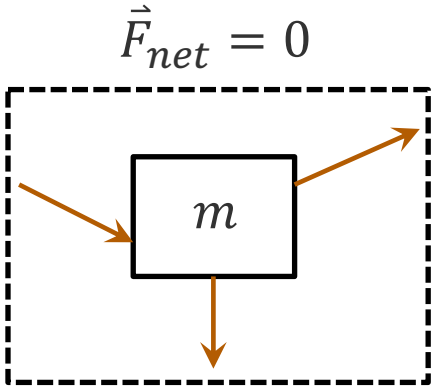
Newton's Third Law

Interacting bodies apply equal (in magnitude) and opposite (in direction) forces on each other

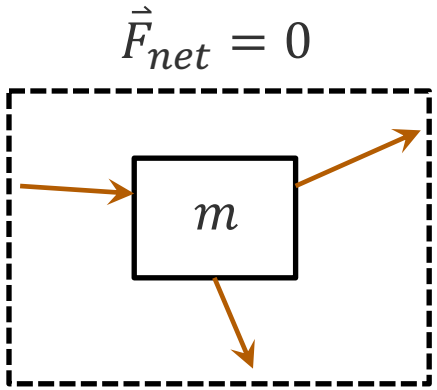
Newton's Laws of Motion

Newton's First Law:

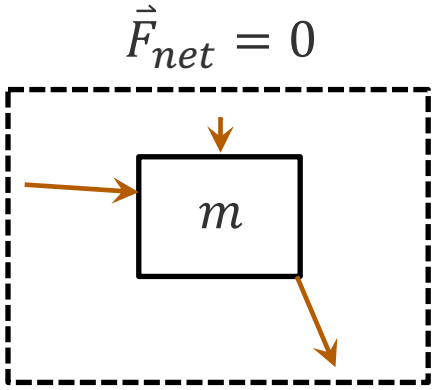
The "motional" status of an object stays the same if no net external forces is acting on it.



Object at rest



Object moves with constant velocity

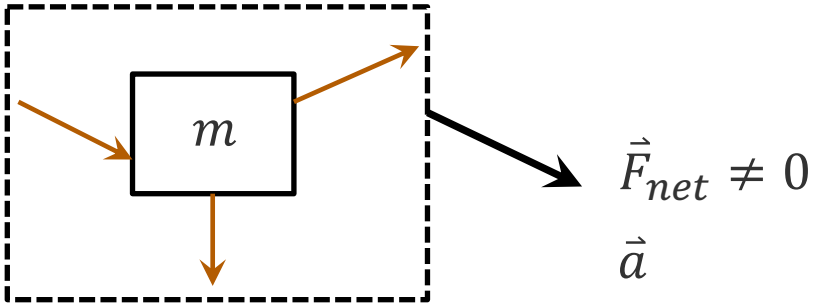


Object moves in a certain direction

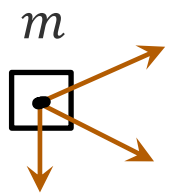
Newton's Laws of Motion

Newton's Second Law:

If there is a net external force acting on an object, it will accelerate.



Object accelerates



The free-body Diagram

1) $\vec{F}_{net} = m\vec{a}$

Units of force is Newton
N ≡ kg·m/s²

2) Acceleration is in the direction as net force.

3) The object is in equilibrium when $\vec{a} = 0$.

4) We can split multi-dimensional motion into

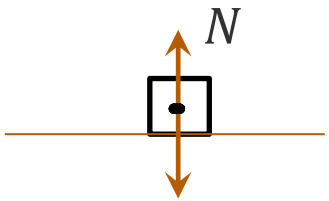
$F_{net_x} = ma_x$
$F_{net_y} = ma_y$
$F_{net_z} = ma_z$

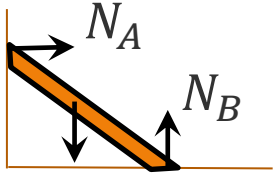
Types of Mechanical Forces

1) Gravitational force. Always directed vertically downward. $F = mg$.

2) The weight of an object is the force it pushes on a flat surface. Not necessarily equal to mg. Example: object on free-fall has zero weight.

3) The normal force (N) is the force that a fixed (not necessarily stationary) surface pushes back at an object resting against it.





4) Friction is the force an object is subjected to when it slides on a rough surface.

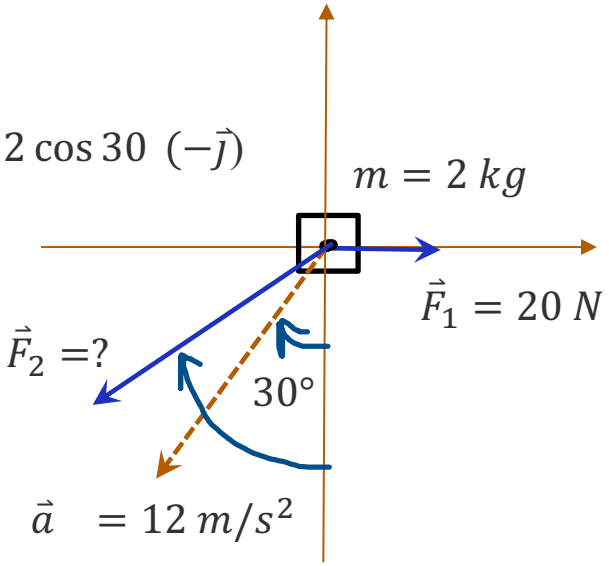
5) Tension is the force created in a connecting wire/rope/cable/cord (a translational force).

6) Drag is the force an object is subjected to when it moves in a viscous medium (fluids/gases).

Examples

1) Find the second force and the angle.

Put in vector form. $\vec{F}_1 = 20\,\vec{\imath}$ $\vec{a} = 12\sin 30\ (-\vec{\imath}) + 12\cos 30\ (-\vec{\jmath})$



$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 = m\,\vec{a} \qquad \vec{F}_2 = m\,\vec{a} - \vec{F}_1$$

$$\vec{F}_2 = (2)(-12\sin 30\ \vec{\imath} - 12\cos 30\ \vec{\jmath}) - 20\,\vec{\imath}$$

$$\vec{F}_2 = -32.0\,\vec{\imath} - 20.8\,\vec{\jmath}\ \ N \qquad angle = \tan^{-1}\frac{32}{20.8} = 57°$$

2) An object of mass 2 kg is subjected to a variable force such that its position at any time is given by $x(t) = 3 + 4t + ct^2 - 2t^3$, where c is a constant. At , $t = 3$ s, $\vec{F}_1 = -36\,\vec{\imath}$ N. Find c.

$$\frac{d^2x}{dt^2} = 2c - 12t$$

$$\left.\frac{d^2x}{dt^2}\right|_{t=3} = 2c - 36$$

At any time, $\vec{F}_{net} = m\,\vec{a}$

At $t = 3$ s, $-36\,\vec{\imath} = (2)\left.\frac{d^2x}{dt^2}\right|_{t=3}\vec{\imath}$ → $-36 = (2)(2c - 36)$ $\qquad c = 9\ m/s^2$

Q & A

Send me an email if
you have any
questions.